Sanjay Suresh

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Education

2026– **University of Wisconsin–Madison**
M.S. in Physics: Quantum Computing (MSPQC)
- **Select Coursework**: Quantum Computing Laboratory, Qubit Tune-Up and Programming, Solid State, Graduate Seminar on Quantum Computing

2022–2026 **University of Wisconsin–Madison**
B.S. in Computer Science and Mathematics; Physics Certificate
- **Select Coursework**: Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Algorithms, Real Analysis, Abstract Algebra

Research

2025– **Distributed Quantum Computing**
Argonne National Laboratory, advised by Dr. Rajkumar Kettimuthu
- Designed and built **SeQUeNCe.DQC**, a top-level module turning SeQUeNCe (**2026 R&D 100 Winner**) into a full-stack DQC simulator: circuit-builder, compiler, central controller, and per-node worker agents.
- Implemented DQC nodes with separate data and communication memory and TeleGate/TeleData teleportation primitives.
- Built QAP, random, and time-sliced FGP-rOEE compilers behind a compiler-agnostic interface, plus ket-vector and density-matrix simulation backends; reproduced the 71% Grover success rate of a *Nature* trapped-ion experiment. Presented at the 2025 Chicago Quantum Summit.

2024– **Fault-Tolerant Quantum Computing**
University of Wisconsin–Madison, advised by Prof. Swamit Tannu
- Developed benchmark suites for fault-tolerant quantum hardware; used variational autoencoders to learn latent-space representations of quantum circuits, enabling similarity analysis based on metrics such as T-gate density.
- Currently developing compilation strategies for the Tour de Gross architecture.

2026– **Measurement-Based Quantum Computing**
University of Oxford (remote collaboration), advised by Dr. Sathya Subramanian
- Extending graph state verification protocols to more sophisticated error models for fault-tolerant measurement-based quantum computing.

2023– **Quantum Algorithms for Scientific Computing**
University of Wisconsin–Madison, advised by Prof. Krishnan Suresh
- Formulated sparse approximate inverse (SPAI) preconditioner construction as a QUBO and computed it on D-Wave Advantage hardware to accelerate conjugate gradient solves of poorly conditioned linear systems.
- Reduced the box-algorithm contraction ratio for **30–40%** faster solution of random linear systems, and adapted annealing to inequality-constrained portfolio optimization. Presented a talk at WCCM 2024.
- Developed an open-source library for structured state-preparation with applications to engineering. Presented a poster and co-taught a workshop at QC 2026.

Publications

SeQUeNCe.DQC: Full-stack Simulation of Distributed Quantum Computing. S. Suresh, C. Zhan, J. Chung, R. Kettimuthu. Submitted to *IEEE ICC 2027*.
PyEncode: An Open-Source Library for Structured Quantum State Preparation. K. Suresh, S. Suresh. *Quantum Sci. Technol.*, 2026.
Computing a Sparse Approximate Inverse on Quantum Annealing Machines. S. Suresh, K. Suresh. *JCISE*, 2025.
Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing. S. Suresh, K. Suresh. *Eng. Optimization*, 2024.

Projects

SurfaceScope (in progress). Building an interactive tool to teach quantum error correction: visualizes Stim-sampled errors, syndrome extraction, and decoding on a surface code, step by step; expanding toward support for more codes. *Python, Flask, React, TypeScript, Stim, PyMatching.*
Studia. Full-stack AI tutoring platform with structured lessons, PDF context extraction via RAG, and real-time Manim animations. Deployed in CS 639 (Prof. Swamit Tannu). *Next.js, TypeScript, Convex, Vercel AI SDK, Manim.*

Skills, Awards & Outreach

Quantum & Languages: Qiskit, Stim, PyMatching, PsiQuantum Workbench, SeQUeNCe, pytket, PennyLane, QuTiP, D-Wave Ocean; Python, TypeScript, C, Java, Haskell
Atlas Fellowship (2022): Selected from **7,000+ applicants** worldwide (**114 recipients**) for a $50K fellowship.
Printable Programming (2019 – Present): Authored *A Visual Approach to Programming for Middle and High School Students* (ISBN: 979-8470081841); taught CAD at the Boys and Girls Club of Dane County.
Carnatic Violin: Professional violinist and accompanist to frontline artists at major festivals across India and the US.